GESTURE-CONTROLLED ROBOT CAR

Project Documentation

Note: The uploaded specification references two project images (robot photo and wiring diagram), but those image files were not included in this chat. This PDF contains the complete documentation text and is ready for those images to be inserted when available.

1. Project Overview

A gesture-controlled robot car using an AI-Thinker ESP32-CAM as the main controller. The ESP32-CAM interprets hand gestures, controls one DC motor through an L298N motor driver, drives a servo on GPIO13, and provides a live camera view for remote operation.

2. Objectives

Gesture-based control, ESP32-CAM programming, motor control, servo positioning, wireless robotics, remote camera viewing, and power management.

3. Components

AI-Thinker ESP32-CAM, L298N motor driver, one DC motor, servo motor, battery, DC-DC buck converter, chassis, wheels, jumper wires, USB-to-Serial programmer, and a computer.

4. System Flow

Hand Gesture → ESP32-CAM → Control Signal → L298N → DC Motor. The servo is controlled by GPIO13 and the camera provides visual feedback.

5. Exact Wiring

GPIO14 → IN1, GPIO15 → IN2, GPIO13 → Servo Signal, OUT1/OUT2 → Single DC Motor, Battery + → L298N 12V, L298N 12V → Buck IN+, Buck OUT+ → ESP32 5V, Battery–/ESP32 GND/Servo GND → L298N GND.

6. Power System

The battery powers the L298N. A buck converter regulates voltage for the ESP32-CAM. Verify the buck converter output before connecting the ESP32-CAM 5V pin.

7. Build Sequence

Assemble chassis, mount motor, wheel, L298N, ESP32-CAM, servo, battery, buck converter, complete wiring, verify connections, upload code, and test.

8. Programming

Use Arduino IDE, install the ESP32 board package, select AI-Thinker ESP32-CAM, connect a USB-to-Serial adapter, choose the correct COM port, compile, upload, reset, and test.

9. Testing

Test power, ESP32 startup, servo, motor, gesture control, camera, then perform a full-system validation.

10. Safety

Disconnect power before rewiring, check polarity, avoid shorts, verify buck voltage, secure wiring, and stop testing if components overheat.

Troubleshooting

Problem	What to Check
ESP32 will not upload	Programming mode, COM port, USB-to-Serial wiring
Servo does not move	GPIO13 signal, 5V supply, common ground
Motor does not move	GPIO14/15 wiring, OUT1/OUT2, battery voltage
Camera not working	ESP32-CAM power and firmware configuration
ESP32 resets	Buck converter output and battery stability

GitHub Repository Structure

```
gesture-controlled-car/
■■■ images/
■ ■■■ robot.jpg
■ ■■■ circuit-diagram.png
■■■ code/
■ ■■■ car-controller.ino
■■■ documentation/
■ ■■■ gesture-controlled-car-documentation.pdf
■■■ README.md
```

Future Improvements

Add a second motor, improve gesture recognition, expand gesture commands, improve camera streaming, add pan/tilt, obstacle detection, and optimize power management.